A Smooth Distributed Formation Control Method for Quadrotor UAVs Under Event-Triggering Mechanism and Switching Topologies

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***Abstract*—This paper focuses on the formation control problem of a team of quadrotor unmanned aerial vehicles (UAVs) subject to switching directed topologies. We present a distributed event-triggered formation control algorithm via hierarchical framework, which achieves the formation control objective with weak assumptions on switching topologies. In particular, we develop a distributed control force that enables the achievement of the target formation, which only relies on the neighbors' position information under event-triggering mechanism. The proposed algorithm can maintain smooth characteristics when confronted with discontinuities stemming from event-triggering mechanism and switching interactions among agents. Additionally, the boundedness of the command control force can be guaranteed by employing an auxiliary system. Subsequently, we develop an attitude tracking control algorithm to guarantee that the attitude error almost globally exponentially converges to zero. Finally, a numerical example is carried out to validate the theoretical results.**

***Index Terms*—Formation control, switching topologies, event-triggering mechanism, unmanned aerial vehicles (UAVs).**

I. Introduction

COOPERATIVE control of multiple unmanned aerial vehicles (UAVs) has become an active research topic due to its wide applications in surveillance, reconnaissance, and environmental monitoring [1], [2], [3], [4]. In contrast to a single UAV, cooperative control of multiple UAVs offers numerous advantages including reduced costs, greater robustness and reliability, and enhanced efficiency. As a typical class of cooperative control, formation control refers to the technology of enabling multiple UAVs to fly according to predetermined spatial distribution and motion patterns.

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Quadrotor drone is a typical representative of UAVs, which attracts significant interest due to its easy operation, simplicity of structure and superior performance. However, the dynamics model of UAVs presents an underactuated characteristics with four degrees of freedom (DOF) in control inputs to govern six-DOF motion. This complexity poses challenges in designing control algorithms for UAVs. A hierarchical framework is commonly utilized in the control design of quadrotor drone [5], [6], [7], [8] to overcome its underactuated nature. This framework is terms as inner-outer loops control structure. The outer loop is dedicated to position control, while the inner loop governs attitude regulation. Several control approaches have been proposed for individual UAV via the hierarchical framework, including predictive control [9], backstepping control [10], robust control [11] and fault-tolerant control [12].

Drawing from the consensus theory of multi-agent systems, numerous distributed algorithms have emerged to tackle the formation control problem. According to types of sensing capability and control variables, these algorithms are classified into position-based, displacement-based, distance-based and bearing-based [13]. For instance, several displacement-based formation algorithms were proposed in [14], [15], [16] to accomplish prescribed translational motion. In [17], [18], [19], distance-based formation control algorithms were developed to finish specified formation shape and formation maneuver. In [20], [21], [22], [23], according to bearing rigidity theory, some formation algorithms were proposed for multi-agent systems with different dynamics. The formation algorithms described above depend on the essential condition that communication topologies among UAVs are fixed. Due to the communication distance restriction or environmental interference, communication link between neighboring UAVs may be weak or even interrupted. Thus, it is more reasonable to represent the communication relationship among the UAVs with switching topologies rather than fixed ones. In [24], [25], [26], several time-varying formation control algorithms were proposed for swarm UAV systems over directed switching topology. The authors of [27] addressed the issues of uniformly jointly switching topology of multiple UAVs formation. In [28], a distributed formation algorithm via output agreement method is developed for multirotors under directed and switching topology. The authors of [29] addressed the bearing formation control problems with single-integrator dynamics under switching interaction topologies. Note that the switching graphs considered in [24], [25], [26] are required to be connected at each time instant, while [27], [28], [29] relaxes this topology condition. Nevertheless, the developed algorithms in [27], [28], [29] are all discontinuous caused by switching interactions.

To improve resource utilization and ease network load, event-based control method is merged into the design of formation control of multiple UAVs, and some event-triggered formation control algorithms are proposed in [30], [31], [32]. Particularly, in [30], a dynamic event-triggered distributed algorithm was proposed with fixed-time convergence to achieve target encirclement task. In [31], an observer-based formation tracking algorithm via event-triggered mechanism was proposed for multi-UAVs systems subject to replay attacks. In [32], an event-triggered adaptive neural network cooperative algorithm was proposed for six-rotor UAVs to achieve the desired performance within a finite time. Moreover, in [33] and [34], two distributed algorithms were developed to address the formation control problems with output constraints and sensor attacks. Note that the above formation algorithms in [30], [31], [32], [33], [34] are discontinuous caused by event-triggered method. However, how to overcome the discontinuity issues arising from event-triggered method and switching topologies, and develop a smooth formation control algorithm is still a challenging problem.

Inspired by the above discussions, this article proposes a smooth distributed formation control algorithm, which can address the discontinuity issues caused by event-triggered method and switching topologies. The union of the switching graphs among UAVs is assumed to contain a spanning tree over some finite subintervals. Using the hierarchical framework, a distributed event-triggered control force is firstly developed to accomplish desired formation tasks. Then, the desired attitude of an UAV is extracted from the designed control force, which is regarded as the tracking reference command, and a control torque is designed to achieve almost globally exponential stability of the attitude error systems. Compared with the related results, the main contributions of this paper are three-fold.

- c1) Compared with some formation control algorithms under fixed topologies or switching topologies with invariably connected conditions [17], [18], [19], [20], [21], [22], [23], the developed formation algorithm requires weak assumption on the switching topologies, i.e., jointly connected directed switching topologies. The relaxed assumption allows the switching graphs to be disconnected at all times.
- c2) A formation control algorithm is proposed by using a high-order dynamical system to address the discontinuity issues caused by event-triggering mechanism and switching topologies, thereby achieving a smooth control force for each UAV. Moreover, unlike the formation algorithms in [21], [22], [23] that require full state information from the neighbors, the proposed algorithm only needs the neighbors' position information. This nice property makes it easier to be implemented in practical applications.
- c3) The proposed event-triggered formation control algorithm can effectively reduce the communication cost, and exclude the Zeno behavior. In addition, based on a hierarchical framework, an attitude tracking control algorithm is designed by using the modified Rodrigue parameters (MRPs), under which the attitude errors almost exponentially converge to zero.

The rest of this paper is organized as follows. After dispensing with the preliminaries and formation control problem of UAVs in Section II and III, we outline the primary outcomes, encompassing the development of control laws and stability analysis in Section IV. Section V conducts numerical simulations to evaluate the proposed control strategy, followed by conclusions presented in Section VI.

II. Preliminaries

A. Notation

Let I_n be the $n \times n$ identity matrix, and $\mathbf{1}_n$ represents the $n \times 1$ ones vector. $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ denote the maximum and minimum eigenvalues of square matrix. $\otimes$ denotes the Kronecker product, $\text{diag}(x_1, \ldots, x_n)$ denotes the diagonal matrix. $|\cdot|$ and $\|\cdot\|$ denote the absolute value and Euclidean norm, respectively. For any vector $x = [x_1, x_2, x_3]^T \in \mathbb{R}^3$, $x^\times = [0, -x_3, x_2; x_3, 0, -x_1; -x_2, x_1, 0]$.

B. Graph Theory

A switching graph $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t))$ is used to model the interaction of all the UAVs, which $\mathcal{V} = \{1, 2, \ldots, n\}$ and $\mathcal{E}(t) \subseteq \mathcal{V} \times \mathcal{V}$ is a time-varying set of edges. Let $\mathcal{G}_p = \{\mathcal{G}_1, \mathcal{G}_2, \cdots\}$ be the set of all possible directed graphs. Suppose that the graph $\mathcal{G}(t)$ switches among the components of $\mathcal{G}_p$ at a sequence of time instants t_k, $k = 1, 2, \cdots$ such that $\inf_k(t_{k+1} - t_k) \geq t_d > 0$, where t_d is the dwell time. The union graph of $\mathcal{G}$ over the time interval $[t_1, t_2)$ is denoted as $\bigcup_{[t_1,t_2)} \mathcal{G}(t) = (\mathcal{V}, \bigcup_{[t_1,t_2)} \mathcal{E}(t))$. $\mathcal{N}_i(t) = \{j : (j, i) \in \mathcal{E}(t)\}$ is the neighbor set of i at time t, and $\mathcal{A}(t) = [a_{ij}(t)] \in \mathbb{R}^{n\times n}$ is the adjacency matrix of $\mathcal{G}(t)$ with $a_{ij}(t) = 1$ if $(j, i) \in \mathcal{E}(t)$ and $a_{ij}(t) = 0$ otherwise. The Laplacian matrix $L(t) = [l_{ij}(t)] \in \mathbb{R}^{n\times n}$ of $\mathcal{G}(t)$ is defined as $l_{ii}(t) = \sum_{j\neq i} a_{ij}(t)$ and $l_{ij}(t) = -a_{ij}(t)$ for $i \neq j$.

Assumption 1: The dwell time t_d is a positive constant and there exists a positive constant $T > 0$ such that $\bigcup_{[t,t+T)} \mathcal{G}(t)$ contains a directed spanning tree for $\forall t \geq 0$. The graph $\mathcal{G}(t)$ is termed as jointly connected directed switching topologies.

III. Problem Statement

For a group of n quadrotor drones, the node set $\mathcal{V} = \{1, 2, \ldots, n\}$ can be used to label the set of UAVs. There is no leader in the group and each UAV flies autonomously. The motion of each UAV is modeled in the inertia frame $\mathcal{F}_I = \{O_I x_I y_I z_I\}$ and body-fixed frame $\mathcal{F}_{bi} = \{O_{bi} x_{bi} y_{bi} z_{bi}\}$. We apply the modified Rodrigue parameters (MRPs) $\sigma_i = \hat{k}_i \tan(\frac{\phi_i}{4})$ to describe the attitude of ith UAV, where $\phi_i \in (-2\pi, 2\pi)$ is the Euler angle and $\hat{k}_i$ is the Euler axis.

For $\forall\, i \in \mathcal{V}$, its kinematics and dynamics are described as

$$\dot{p}_i = v_i, \tag{1a}$$

$$\dot{v}_i = -g\hat{e}_3 + \frac{T_i}{m_i}R(\sigma_i)\hat{e}_3, \tag{1b}$$

$$\dot{\sigma}_i = G(\sigma_i)\omega_i, \tag{1c}$$

$$J_i\dot{\omega}_i = -{\omega_i}^{\times}J_i\omega_i + \tau_i, \tag{1d}$$

where $p_i = [{p_i}^x, {p_i}^y, {p_i}^z]^T \in \mathbb{R}^3$ and $v_i = [{v_i}^x, {v_i}^y, {v_i}^z]^T \in \mathbb{R}^3$ represent the ith UAV's position and velocity in frame $\mathcal{F}_I$, T_i is the thrust magnitude, m_i is mass of UAV, $\hat{e}_3 = [0, 0, 1]^T$, and g is the gravity acceleration. $R(\sigma_i) = I_3 + 4\frac{1-\sigma_i^T\sigma_i}{(1+\sigma_i^T\sigma_i)^2}\sigma_i^{\times} + \frac{8\sigma_i^{\times}\sigma_i^{\times}}{(1+\sigma_i^T\sigma_i)^2}$ is the rotation matrix from frame $\mathcal{F}_{bi}$ to frame $\mathcal{F}_I$. $J_i \in \mathbb{R}^{3\times 3}$ is the inertia matrix, $\omega_i = [\omega_i^x, \omega_i^y, \omega_i^z]^T$ is the angular velocity, $\tau_i = [\tau_i^x, \tau_i^y, \tau_i^z]^T$ is the control torque of the ith UAV expressed in $\mathcal{F}_{bi}$ and $G(\sigma_i) = \frac{1}{4}((1-\sigma_i^T\sigma_i)I_3 + 2\sigma_i^{\times} + 2\sigma_i\sigma_i^T)$. Some properties of MRPs are shown as [35]

$$\sigma_i^T G(\sigma_i)\omega_i = \left(\frac{1+\sigma_i^T\sigma_i}{4}\right)\sigma_i^T\omega_i, \tag{2a}$$

$$\|G(\sigma_i)\| = \left(\frac{1+\sigma_i^T\sigma_i}{4}\right)^2 I_3. \tag{2b}$$

The goal of this paper is to develop a distributed control strategy for multiple UAVs under switching topologies such that the UAVs group can maintain a desired formation configuration while following a time-varying reference velocity. More specifically, setting a desired position offset $\delta_{ij} = \delta_i - \delta_j$ between i and j, and a reference velocity $v_r(t)$, the formation control objective can be accomplished if the following conditions are satisfied

$$\lim_{t\to\infty} p_i(t) - p_j(t) = \delta_{ij}, \lim_{t\to\infty} v_i(t) - v_r(t) = 0. \tag{3}$$

IV. Main Results

A distributed control strategy via a hierarchical framework is proposed to accomplish the formation objective (3). Firstly, we transform equivalently the formation control problem into two subproblems, and then a distributed control force is developed under weak assumption on switching graph. Afterwards, the desired attitude of an UAV is extracted from the designed control force, which is regarded as the tracking reference command, and then a control torque is designed to achieve almost globally exponential stability of the attitude error systems.

A. Problem Transformation

Let $p_i^e = p_i - \int_0^t v_r(s)\mathrm{d}s - \delta_i$ and $v_i^e = v_i - v_r\ \forall i \in \mathcal{V}$ be the position and velocity error, where δ_i is the desired position offset with respect to the formation center. Moreover, we know that $p_i^e - p_j^e \to 0$ is equivalent to $p_i - p_j \to \delta_{ij}$. Substituting p_i^e and v_i^e into the dynamical model (1) yields

$$\dot{p}_i^e = v_i^e, \tag{4a}$$

$$\dot{v}_i^e = -g\hat{e}_3 + \frac{T_i}{m_i}(R(\sigma_i) - R(\sigma_i^c))\hat{e}_3 + u_i, \tag{4b}$$

where $u_i = \frac{T_i}{m_i}R(\sigma_i^c)\hat{e}_3$ represents the command force to be determined later and σ_i^c denotes the desired attitude. In terms of $\|R(\sigma_i^c)\hat{e}_3\| = 1$, the applied thrust is obtained as $T_i = m_i\|u_i\|, i \in \mathcal{V}$. In addition, using the hierarchical framework, the desired attitude is obtained in the following lemma.

Lemma 1 ([5]): Suppose that the command force $u_i = [u_i^x, u_i^y, u_i^z]^T$ satisfies $u_i \notin \mathbb{U} = \{u \in \mathbb{R}^3 \mid u = [0, 0, u_z]^T, u_z \le 0\}$. The desired attitude σ_i^c and angular velocity ω_i^c can be extracted as

$$\sigma_i^c = \frac{m_i}{2(1+q_{i0})T_i q_{i0}}\begin{bmatrix} -u_i^y \\ u_i^x \\ 0 \end{bmatrix}, \tag{5}$$

$$\omega_i^c = \frac{1}{\gamma_{i,1}^2\gamma_{i,2}}\Xi(u_i)\dot{u}_i,$$

$$\Xi(u_i) = \begin{bmatrix} -u_i^x u_i^y & -(u_i^y)^2 + \gamma_{i,1}\gamma_{i,2} & u_i^y\gamma_{i,2} \\ (u_i^x)^2 - \gamma_{i,1}\gamma_{i,2} & u_i^x u_i^y & -u_i^x\gamma_{i,2} \\ u_i^y\gamma_{i,1} & -u_i^x\gamma_{i,1} & 0 \end{bmatrix}, \tag{6}$$

where $q_{i0} = \sqrt{\frac{1}{2} + \frac{m_i u_i^z}{2T_i}}$, $\gamma_{i,1} = (T_i/m_i)$, $\gamma_{i,2} = \gamma_{i,1} + (g/u_i^z)$.

With the desired attitude σ_i^c, we define attitude error σ_i^e as

$$\sigma_i^e = \frac{\sigma_i^c(\sigma_i^T\sigma_i - 1) + \sigma_i(1 - \|\sigma_i^c\|^2) - 2(\sigma_i^c)^{\times}\sigma_i}{1 + \|\sigma_i^c\|^2\|\sigma_i\|^2 + 2(\sigma_i^c)^T\sigma_i}.$$

The dynamics of σ_i^e is given by

$$\dot{\sigma}_i^e = G(\sigma_i^e)\omega_i^e, \omega_i^e = \omega_i - R^T(\sigma_i^e)\omega_i^c, \tag{7}$$

where ω_i^e is the angular velocity error. Based on (1) and (7), the dynamics of ω_i^e is

$$J_i\dot{\omega}_i^e = -{\omega_i}^{\times}J_i\omega_i + \tau_i + J_i((\omega_i^e)^{\times}R^T(\sigma_i^e)\omega_i^c - R^T(\sigma_i^e)\dot{\omega}_i^c). \tag{8}$$

The formation objective (3) is achieved if the following conditions are satisfied

$$\lim_{t\to\infty} p_i^e(t) - p_j^e(t) = 0, \lim_{t\to\infty} v_i^e(t) = 0, \tag{9}$$

$$\lim_{t\to\infty} \sigma_i^e(t) = 0, \lim_{t\to\infty} \omega_i^e(t) = 0. \tag{10}$$

B. Distributed Control Force Design

This section presents a distributed formation command force u_i such that $\lim_{t\to\infty} p_i^e(t) - p_j^e(t) = 0$ and $\lim_{t\to\infty} v_i^e(t) = 0$ can be satisfied. To guarantee the boundedness of the thrust T_i, we define the virtual position and velocity errors $\bar{p}_i^e = p_i^e - \eta_i$ and $\bar{v}_i^e = v_i^e - \dot{\eta}_i$ for $i \in \mathcal{V}$, where $\eta_i \in \mathbb{R}^3$ is an auxiliary variable. Based on (8), we can get

$$\dot{\bar{p}}_i^e = \bar{v}_i^e, \tag{11a}$$

$$\dot{\bar{v}}_i^e = -g\hat{e}_3 + u_i - \dot{v}_r - \ddot{\eta}_i + \frac{T_i}{m_i}R(\sigma_i^d)[R(\sigma_i^e) - I_3])\hat{e}_3. \tag{11b}$$

The desired formation configuration is achieved when $\lim_{t\to\infty} \bar{p}_i^e(t) - \bar{p}_j^e(t) = 0$, $\lim_{t\to\infty} \bar{v}_i^e(t) = 0$ and $\lim_{t\to\infty} \eta_i(t) = 0$, $\lim_{t\to\infty} \dot{\eta}_i(t) = 0$, $\forall i, j \in \mathcal{V}$. To this end, we develop a distributed control algorithm incorporating second-order auxiliary system

$$u_i = \dot{v}_r + g\hat{e}_3 - k_\eta[\tanh(\eta_i) + \tanh(\dot{\eta}_i)], \tag{12a}$$

$$\ddot{\eta}_i = -k_\eta[\tanh(\eta_i) + \tanh(\dot{\eta}_i)] + \theta_i, \tag{12b}$$

$$\theta_i = k_\theta(\dot{\bar{p}}_i^e - q_i) - \dot{q}_i, \tag{12c}$$

where k_η, k_θ are positive constants, η_i and $\theta_i \in \mathbb{R}^3$ are the state and input of second-order auxiliary system (12b). The variable q_i in (12c) is defined as $q_i = -\gamma_i(\bar{p}_i - y_i)$, where $\gamma_i > 0$ and $\bar{p}_i = p_i - \eta_i$, and $y_i \in \mathbb{R}^3$ is the output of the following high-order dynamical system

$$\dot{x}_{h,i} = \lambda_{h,i}\left(x_{h+1,i} - x_{h,i}\right) + v_r, \tag{13a}$$

$$\dot{x}_{k,i} = -\lambda_{k,i}\sum_{j=1}^{n} a_{ij}(t)x_{k,i} + \lambda_{k,i}d_i + v_r, \tag{13b}$$

$$y_i = x_{1,i}. \tag{13c}$$

where $x_{h,i}, x_{i,k} \in \mathbb{R}^3, h = 1, 2, \ldots, k-1$ are the system states, $k \geq 2$ is the order of this system, $\lambda_{h,i}$ and $\lambda_{k,i}$ are positive constants for $\forall i \in \mathcal{V}$, and $d_i \in \mathbb{R}^3$ is the input of system (13) and is designed as

$$d_i = \sum_{j=1}^{n} a_{ij}(t)\left(\hat{\bar{p}}_j(t) + \int_{t_k^j}^{t} v_r(s)\mathrm{d}s + \delta_{ij}\right). \tag{14}$$

where $\hat{\bar{p}}_j(t) = \bar{p}_j(t_k^j), t \in [t_k^j, t_{k+1}^j)$ denotes the broadcast state of agent j and $\{t_k^j\}_{k=1}^{\infty}$ is the triggering time sequence of agent j.

We introduce the even-triggered communication mechanism to avoid the consistent communication. For agent $i \in \mathcal{V}$, the broadcast state $\hat{\bar{p}}_i(t)$ that is transmitted to its neighbors is defined as

$$\hat{\bar{p}}_i(t) = \bar{p}_i\left(t_k^i\right), \quad t \in \left[t_k^i, t_{k+1}^i\right). \tag{15}$$

where t_k^i denotes the kth triggering time of agent i. The triggering time sequence $\{t_k^i\}$ is determined by the following updated rule

$$t_{k+1}^i = \left\{ t > t_k^i \,\middle|\, \| \bar{p}_i(t) - \hat{\bar{p}}_i(t) - \int_{t_k^i}^{t} v_r(s)\mathrm{d}s \| \geq c_0 \mathrm{e}^{-\alpha t} \right\}, \tag{16}$$

with the positive constants $\alpha > 0$ and $c_0 > 0$.

Let $x_i = [x_{1,i}^T, x_{2,i}^T, \ldots, x_{k,i}^T]^T \in \mathbb{R}^{3k}$, and the compact form of above high-order dynamical system (13) is obtained as

$$\begin{aligned} \dot{x}_i &= (A_i \otimes I_3)\, x_i + (B_i \otimes I_3)\, d_i + (E_i \otimes I_3)\, v_r, \\ y_i &= (C_i \otimes I_3)\, x_i, \end{aligned} \tag{17}$$

where the matrices $A_i \in \mathbb{R}^{k\times k}$, $B_i \in \mathbb{R}^{k}$ and $C_i \in \mathbb{R}^{1\times k}$ have the following form

$$A_i = \begin{bmatrix} -\lambda_{1,i} & \lambda_{1,i} & 0 & \ldots & 0 & 0 \\ 0 & -\lambda_{2,i} & \lambda_{2,i} & \ldots & 0 & 0 \\ \vdots & & \ddots & & \ldots & \vdots \\ \vdots & & & \ddots & \ldots & \vdots \\ 0 & 0 & 0 & \ldots & -\lambda_{k-1,i} & \lambda_{k-1,i} \\ 0 & 0 & 0 & \ldots & 0 & -\lambda_{k,i}\sum\limits_{j=1}^{n} a_{ij}(t) \end{bmatrix},$$

$$B_i = [0, 0, \ldots, 0, \lambda_{k,i}]^T, C_i = [1, 0, \ldots, 0, 0]^T,$$

$$E_i = [1, 1, \ldots, 1, 1]^T.$$

The second-order auxiliary system (12b) is to guarantee the boundedness of the control force u_i such that the applied thrust T_i is bounded. From the expression of (12a) and $\| \tanh(\cdot)\| \leq 1$, it follows that the command force u_i is bounded by $\|u_i\| \leq g + 2k_\eta + \sup_{t\geq 0}\{\|\dot{v}_r(t)\|\}$. This implies that T_i is bounded with $T_i \leq m_i(g + \sup_t\{\|\dot{v}_r(t)\|\} + 2k_\eta)$. Note that the upper bound of T_i is irrelevant to the system states and is determined only by the gravity acceleration g and the control gain parameter k_η.

From the extracted desired attitude σ_i^c and angular velocity ω_i^c in Lemma 1, we know that angular velocity ω_i^c and its derivative $\dot{\omega}_i^c$ involve the first-order and second-order derivative of formation controller u_i. However, the typical formation controller $u_i = -\sum_{j=1}^{N} a_{ij}(t)(p_i(t) - p_j(t) - \delta_{ij}) - \sum_{j=1}^{N} a_{ij}(t)(v_i(t) - v_j(t))$ in [13] and [14] does not satisfy the continuity property when event-triggering mechanism and switching topologies are considered. To deal with this issue, we introduce the high-order dynamical system (13) to ensure the differentiability and smoothness of formation controller u_i, which can address the discontinuity issues caused by event-triggering mechanism and switching topologies. In particular, when we choose $k = 2$ of system (13), a second-order dynamical system is obtained as

$$\dot{x}_{1,i} = \lambda_{1,i}\left(x_{2,i} - x_{1,i}\right) + v_r, \tag{18a}$$

$$\dot{x}_{2,i} = -\lambda_{2,i}\sum_{j=1}^{n} a_{ij}(t)x_{2,i} + \lambda_{2,i}d_i + v_r, \tag{18b}$$

$$y_i = x_{1,i}. \tag{18c}$$

where d_i has the same form as (14). Under the above second-order dynamical system (18), we can verify that the formation controller u_i and its first-order derivative are both continuous and differentiable. This implies that the developed formation algorithm (12) owns smooth characteristics.

Remark 1: The proposed distributed control scheme (12) only relies on relative position information from neighbors and reference velocity information. Additionally, the use of the high-order dynamical system (13) ensures continuous differentiability of the output y_i such that q_i and control force u_i are also continuously differentiable, even the input d_i involves the

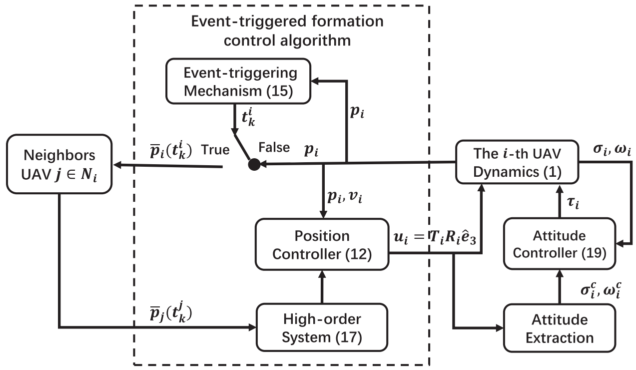

Fig. 1. The structure diagram of the control strategy.

switching information from neighbors. This guarantees a smooth control force for each UAV.

C. Attitude Control Design

We propose an attitude control algorithm to achieve the attitude tracking objective (10). Moreover, the developed algorithm guarantees that the attitude error almost globally exponentially converges to zero.

To achieve $\lim_{t\to\infty} \sigma_i^e(t) = 0$ and $\lim_{t\to\infty} \omega_i^e(t) = 0$ exponentially, the applied torque τ_i is proposed as

$$\tau = -\frac{k_1}{4}(1 + (\sigma_i^e)^T \sigma_i^e)\omega_i^e - k_2(\sigma_i^e + \omega_i^e) + \omega_i^\times J_i \omega_i - J_i((\omega_i^e)^\times R^T(\sigma_i^e)\omega_i^c - R^T(\sigma_i^e)\dot{\omega}_i^c), \tag{19}$$

where k_1 and k_2 are positive control parameters. Substitute (19) into (8) yields

$$J_i \dot{\omega}_i^e = -\frac{k_1}{4}(1 + (\sigma_i^e)^T \sigma_i^e)\omega_i^e - k_2(\sigma_i^e + \omega_i^e). \tag{20}$$

From (19), we know that the designed torque τ_i needs the information of ω_i^c and its derivative $\dot{\omega}_i^c$. The exact expressions of ω_i^c and $\dot{\omega}_i^c$ are given in [5, Lemma 1].

To understand the above control design clearly, we present the structure diagram of the distributed formation control strategy in the following Fig. 1.

D. Stable Analysis

The stabilities of the control force (12) with the event-triggering mechanism (16) and applied torque (19) are discussed in this section. Firstly, the proof of the attitude system stabilization is provided.

Proposition 1: Consider the attitude error dynamical system (7) and (8) under Assumptions 1. The proposed control torque (20) guarantees that attitude error σ_i^e and angular velocity error ω_i^e almost globally exponentially converge to zero.

Proof: Choose the Lyapunov candidate function

$$V_i(\sigma_i^e, \omega_i^e) = \frac{k_1}{2}(\sigma_i^e)^T \sigma_i^e + \frac{1}{2}(\omega_i^e)^T J_i \omega_i^e + (\sigma_i^e)^T J_i \omega_i^e.$$

Note that $\frac{1}{2}\lambda_{\min}(\Phi_i)\|\sigma_i^e + \omega_i^e\|^2 \le V_i(\sigma_i^e, \omega_i^e) \le \frac{1}{2}\lambda_{\max}(\Phi_i)\|\sigma_i^e + \omega_i^e\|^2$, where $\Phi_i = \begin{bmatrix} k_1 I_3 & J_i \\ J_i & J_i \end{bmatrix}$ is a positive definite matrix if $k_1 > \lambda_{\max}(J_i)$.

Based on (1) and (20), the derivative of $V_i(\sigma_i^e, \omega_i^e)$ is obtained as

$$\begin{aligned}
\dot{V}_i &= k_1(\sigma_i^e)^T G(\sigma_i^e)\omega_i^e + (\omega_i^e)^T J_i \dot{\omega}_i^e + (\omega_i^e)^T J_i G(\sigma_i^e)\omega_i^e \\
&\quad + (\sigma_i^e)^T J_i \dot{\omega}_i^e \\
&= \frac{k_1}{4}(1 + (\sigma_i^e)^T \sigma_i^e)\sigma_{ie}^T \omega_i^e + (\omega_i^e + \sigma_i^e)^T J_i \dot{\omega}_i^e \\
&\quad + (\omega_i^e)^T J_i G(\sigma_i^e)\omega_i^e \\
&= \frac{k_1}{4}(1 + (\sigma_i^e)^T \sigma_i^e)\sigma_{ie}^T \omega_i^e + (\omega_i^e + \sigma_i^e)^T(-k_2(\sigma_i^e + \omega_i^e) \\
&\quad - \frac{k_1}{4}(1 + (\sigma_i^e)^T \sigma_i^e)\omega_i^e) + (\omega_i^e)^T J_i G(\sigma_i^e)\omega_i^e \\
&= -k_2(\omega_i^e + \sigma_i^e)^T(\sigma_i^e + \omega_i^e) \\
&\quad - \frac{k_1}{4}(\omega_i^e)^T(1 + (\sigma_i^e)^T \sigma_i^e)\omega_i^e + (\omega_i^e)^T \|J_i\| \|G(\sigma_i^e)\| \omega_i^e \\
&\le -k_2\|\sigma_i^e + \omega_i^e\|^2 \\
&\quad - \frac{(k_1 - \lambda_{\max}(J_i))}{4}(1 + (\sigma_i^e)^T \sigma_i^e)\|\omega_i^e\|^2,
\end{aligned}$$

where the second equation and last equation are obtained by using (2a) and(2b), respectively. It then follows that

$$\dot{V}_i(\sigma_i^e, \omega_i^e) \le -\frac{2k_2}{\lambda_{\max}(\Phi_i)} V_i(\sigma_i^e, \omega_i^e), \tag{21}$$

which implies that $\lim_{t\to\infty} \sigma_i^e(t) = 0$ and $\lim_{t\to\infty} \omega_i^e(t) = 0$ exponentially. □

Remark 2: Unlike the attitude controller in [27], the proposed control torque (19) not only can ensure that the attitude error exponentially converges to zero, but also can achieve almost globally stable for any initial state $\sigma_i^e(0)$ and $\omega_i^e(0)$ except for singular points $\phi_i = \pm 2k\pi$.

We next show the convergence results of $\bar{p}_i^e(t) - \bar{p}_j^e(t)$ and $\bar{v}_i^e$ in the following proposition.

Proposition 2: Consider the error dynamic system (11) under Assumption 1. Provided that $\lim_{t\to\infty} \sigma_i^e(t) = 0$ in Proposition 1 and the gain parameters satisfying $\alpha < \kappa_1$ and $k_\theta \ge \alpha$, the developed control force (12) with event-triggering rule (16) guarantees that $\lim_{t\to\infty} \bar{p}_i^e(t) - \bar{p}_j^e(t) = 0$ and $\lim_{t\to\infty} \bar{v}_i^e(t) = 0$ exponentially.

Proof: Let $s_i = \dot{\bar{p}}_i^e - q_i$. Under the control force (12a), the derivative of s_i can be computed as

$$\begin{aligned}
\dot{s}_i &= \dot{\bar{v}}_i^e - \dot{q}_i \\
&= -g\hat{e}_3 + u_i - \dot{v}_r - \ddot{\eta}_i - \dot{q}_i + \varpi_i \\
&= -\theta_i - \dot{q}_i + \varpi_i \\
&= -k_\theta(\dot{\bar{p}}_i^e - q_i) + \varpi_i \\
&= -k_\theta s_i + \varpi_i,
\end{aligned} \tag{22}$$

where $\varpi_i = \frac{T_i}{m_i} R(\sigma_i^d)[R(\sigma_i^e) - I_3])\hat{e}_3$. Since the thrust magnitude T_i is bounded, $\|R(\sigma_i^e)\| = 1$ and $\|R(\sigma_i^d)\| = 1$, we have that $\|\varpi_i\|$ is bounded. In addition, if $\lim_{t\to\infty} \sigma_i^e(t) = 0$ exponentially, one has that $\lim_{t\to\infty} R(\sigma_i^e(t)) = I_3$ and $\lim_{t\to\infty} \varpi_i(t) = 0$ exponentially. Based on the input-to-state stability in [36], we have that $\lim_{t\to\infty} s_i(t) = 0$ exponentially.

From the high-order dynamical system (13), we can get that

$$y_i = x_{1,i}, \dot{x}_{h,i} = \lambda_{h,i} \left(x_{h+1,i} - x_{h,i}\right) + v_r, \tag{23}$$

for $h = 1, \ldots, k-1$, and

$$\dot{x}_{k,i} = -\lambda_{k,i} \sum_{j=1}^{n} a_{ij}(t) \left(x_{k,i} - \hat{\bar{p}}_j - \int_{t_k^j}^{t} v_r \mathrm{d}s - \delta_{ij} \right) + v_r. \tag{24}$$

Let $\bar{y}_i = y_i - \int_0^t v_r \mathrm{d}s - \delta_i$ and $\bar{x}_{h_i i} = x_{h_i i} - \int_0^t v_r \mathrm{d}s - \delta_i$ for $h = 1, \ldots, k$, It follows that $\bar{y}_i = \bar{x}_{1,i}$ and $q_i = -\gamma_i(\bar{p}_i - y_i) = -\gamma_i(\bar{p}_i^e - \bar{y}_i) = -\gamma_i(\bar{p}_i^e - \bar{x}_{1,i})$. Combined with (23), we have that

$$\begin{aligned}
\dot{\bar{x}}_{h,i} &= \dot{x}_{h,i} - v_r \\
&= \lambda_{h,i} \left(x_{h+1,i} - x_{h,i}\right) + v_r - v_r \\
&= \lambda_{h,i} (x_{h+1,i} - \int_0^t v_r(s)\mathrm{d}s - \delta_i \\
&\quad - x_{h,i} + \int_0^t v_r(s)\mathrm{d}s + \delta_i) \\
&= \lambda_{h,i} \left(\bar{x}_{h+1,i} - \bar{x}_{h,i}\right),
\end{aligned} \tag{25}$$

for $h = 1, \ldots, k-1$. Similarly, we obtain that

$$\dot{\bar{x}}_{k,i} = -\lambda_{ki} \sum_{j=1}^{n} a_{ij} \left(\bar{x}_{k,i} - \hat{\bar{p}}_j(t) - \int_{t_k^j}^{t} v_r \mathrm{d}s + \delta_j + \int_0^t v_r \mathrm{d}s \right).$$

The above equation can be further simplified as

$$\dot{\bar{x}}_{k,i} = -\lambda_{ki} \sum_{j=1}^{n} a_{ij} (\bar{x}_{k,i} - \bar{p}_j^e + e_j), \tag{26}$$

where e_j is given as follows

$$\begin{aligned}
e_j &= \bar{p}_j^e(t) - \hat{\bar{p}}_j(t) - \int_{t_k^j}^{t} v_r \mathrm{d}s + \delta_j + \int_0^t v_r \mathrm{d}s \\
&= \bar{p}_j(t) - \hat{\bar{p}}_j(t) - \int_{t_k^j}^{t} v_r \mathrm{d}s,
\end{aligned}$$

where the last equation is obtained by using $\bar{p}_j^e(t) = \bar{p}_j(t) - \int_0^t v_r(s)ds - \delta_j$.

From $s_i = \dot{\bar{p}}_i^e - q_i$, one has that $\dot{\bar{p}}_i^e = q_i + s_i = -\gamma_i(\bar{p}_i^e - \bar{x}_{i,1}) + s_i$. Define the variable $\zeta = (\zeta_1^\top, \ldots, \zeta_{(k+1)n}^\top)^\top \in \mathbb{R}^{3(k+1)n}$ that satisfies

$$\begin{aligned}
&\zeta_i = \bar{x}_{1,i}, \\
&\vdots \\
&\zeta_{(k-1)n+i} = \bar{x}_{k,i}, \\
&\zeta_{kn+i} = \bar{p}_i^e
\end{aligned} \tag{27}$$

for $i \in \mathcal{V}$. Based on (25) and (26), we obtain the dynamics of state ζ in (27) as

$$\dot{\zeta}_{(h-1)n+i} = -\lambda_{h,i} (\zeta_{(h-1)n+i} - \zeta_{hn+i}), \tag{28a}$$

$$\dot{\zeta}_{(k-1)n+i} = -\lambda_{k,i} \sum_{j=1}^{n} a_{ij}(t) [\zeta_{(k-1)n+i} - \zeta_{kn+j}(t) + \bar{e}_j], \tag{28b}$$

$$\dot{\zeta}_{kn+i} = -\gamma_i (\zeta_{kn+i} - \zeta_i) + s_i, \tag{28c}$$

for $i \in \mathcal{V}$ and $h = 1, \ldots, k-1$. In addition, the above (28) can be rewritten in a compact form

$$\dot{\zeta} = -\left(\tilde{L}(t) \otimes I_3\right) \zeta + \varrho_1(t) + \varrho_2(t), \tag{29}$$

where $\varrho_1(t) = (\mathbf{0}_{3n}^\top, \ldots, \mathbf{0}_{3n}^\top, s^\top)^\top \in \mathbb{R}^{3(k+1)n}$, $s = (s_1^\top, \ldots, s_n^\top)^\top$, $\varrho_2(t) = (\mathbf{0}_{3n}^\top, \ldots, \bar{e}^\top, \mathbf{0}_{3n}^\top)^\top \in \mathbb{R}^{3(k+1)n}$ and $\bar{e} = (e_1^\top, \ldots, e_n^\top)^\top$, and the matrix $\tilde{L}(t)$ is presented as

$$\tilde{L}(t) = \begin{bmatrix}
\Delta_1 & -\Delta_1 & \ldots & 0 & 0 \\
0 & \Delta_2 & \ldots & 0 & 0 \\
\vdots & & \ddots & & \vdots \\
\vdots & & & \ddots & \vdots \\
0 & 0 & \ldots & -\Delta_{k-1} & 0 \\
0 & 0 & \ldots & \Delta_k \mathcal{D}_n(t) & -\Delta_k \mathcal{A}_n(t) \\
-\Gamma & 0 & \ldots & 0 & \Gamma
\end{bmatrix},$$

where $\Gamma = \mathrm{diag}\{\gamma_1, \ldots, \gamma_N\}$ and $\Delta_h := \mathrm{diag}\{\lambda_{h,1}, \ldots, \lambda_{h,N}\}$ for $h = 1, \ldots, k$, and $\mathcal{A}(t)$ is the adjacency matrix and $\mathcal{D}(t)$ denotes in-degree matrix of $\mathcal{G}(t)$.

Inspired by [39], we define the relative error $\tilde{\zeta} = (\zeta_1^\top - \zeta_2^\top, \ldots, \zeta_1^\top - \zeta_{(k+1)n}^\top)$, and its dynamics has following form

$$\dot{\tilde{\zeta}} = -\left(\tilde{L}'(t) \otimes I_3\right) \tilde{\zeta} + \varrho_1(t) + \varrho_2(t). \tag{30}$$

where the elements of $\tilde{L}'(t)$ are denoted by $\tilde{L}'(t)_{ij} = \tilde{L}(t)_{ij} - \tilde{L}(t)_{1j}$ for $i, j = 1, \ldots, N \times (m+1)$.

The system (30) can be regarded as a perturbed system, where $\varrho_1(t) + \varrho_2(t)$ is a perturbed term. The nominal dynamics of this perturbed system is given by the following expression

$$\dot{\tilde{\zeta}} = -\left(\tilde{L}'(t) \otimes I_3\right) \tilde{\zeta}. \tag{31}$$

By following the similar analysis of Theorem 1 in [37], we obtain that the system described by (31) is uniformly exponentially stable under the assumption that the union of the switching graph among UAVs contains a spanning tree. Therefore, the state transition matrix $\Phi(t, \psi), t \geq \psi$ of system (31) satisfies that

$$\|\Phi(t, \psi)\| \leq \beta_1 \, \mathrm{e}^{-\kappa_1 (t-\psi)}, t \geq \psi \geq 0, \tag{32}$$

where $\beta_1 > 0$ and κ_1 is determined by the Laplacian matrix of the union of directed graph $\mathcal{G}(t)$. Under Assumption 1, one has that κ_1 is a positive constant.

The result (32) is essential for subsequent stability analysis. We next continue analyzing the characteristics of system (30).

The solution of (30) is

$$\tilde{\zeta}(t) = \Phi(t,0)\tilde{\zeta}(t)(0) + \int_0^t \Phi(t,\psi)(\varrho_1(\psi) + \varrho_2(\psi))\mathrm{d}\psi. \tag{33}$$

Utilizing the result (32) yields

$$\|\tilde{\zeta}(t)\| \leq \beta_1\|\tilde{\zeta}(0)\|\mathrm{e}^{-\kappa_1 t} + \int_0^t \beta_1\mathrm{e}^{-\kappa_1(t-\psi)}(\|\varrho_1(\psi)\| + \|\varrho_2(\psi)\|)\mathrm{d}\psi. \tag{34}$$

Note that $\|\varrho_1(t)\| = \|s(t)\|$ and $\lim_{t\to\infty} s(t) = 0$ exponentially in Proposition 1, and one further can derive that $\lim_{t\to\infty}\|\varrho_1(t)\| = 0$ exponentially. This implies that $\varrho_1(t)$ satisfies that

$$\|\varrho_1(t)\| \leq \beta_2\mathrm{e}^{-k_\theta t} \quad \forall t \geq 0. \tag{35}$$

where $\beta_2 > 0$. According to the definition of $\varrho_2(t)$ and event-triggering mechanism (16), we obtain that

$$\|\varrho_2(t)\| \leq c_0\sqrt{n}\mathrm{e}^{-\alpha t} \quad \forall t \geq 0. \tag{36}$$

Based on inequalities (35), (36) and $k_\theta > \alpha > 0$, we can get that

$$\|\varrho_1(t)\| + \|\varrho_2(t)\| \leq \bar{c}\mathrm{e}^{-\alpha t} \quad \forall t \geq 0. \tag{37}$$

where $\bar{c}$ is a positive constant. Applying (37) into (34) yields

$$\begin{aligned}\|\tilde{\zeta}(t)\| &\leq \beta_1\|\tilde{\zeta}(0)\|\mathrm{e}^{-\kappa_1 t} + \int_0^t \beta_1\mathrm{e}^{-\kappa_1(t-\psi)}\bar{c}\mathrm{e}^{-\alpha t}\mathrm{d}\psi \\ &\leq \beta_1\|\tilde{\zeta}(0)\|\mathrm{e}^{-\kappa_1 t} + \beta_1\bar{c}\mathrm{e}^{-\kappa_1 t}\int_0^t \mathrm{e}^{(\kappa_1-\alpha)\psi}d\psi \\ &\leq \beta_1\|\tilde{\zeta}(0)\|\mathrm{e}^{-\kappa_1 t} + \frac{\beta_1\bar{c}}{(\kappa_1-\alpha)}(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\kappa_1 t}).\end{aligned}$$

For simplicity, we can express the above equation as follows

$$\|\tilde{\zeta}(t)\| \leq l_1\mathrm{e}^{-\kappa_1 t} + l_2\mathrm{e}^{-\alpha t}, \tag{38}$$

where $l_1 = \beta_1(\|\tilde{\zeta}(t)(0)\| - \frac{\beta_1\bar{c}}{(\kappa_1-\alpha)})$ and $l_2 = \frac{\beta_1\bar{c}}{(\kappa_1-\alpha)}$.

Recalling the definition of $\tilde{\zeta}(t)$ in (30), it follows that

$$\|\bar{p}_i^e(t) - \bar{p}_j^e(t)\| \leq 2(l_1\mathrm{e}^{-\kappa_1 t} + l_2\mathrm{e}^{-\alpha t}),$$

$$\|q_i(t)\| = \gamma_i\|(\bar{p}_i^e(t) - \bar{x}_{1,i}(t))\| \leq 2\gamma_i(l_1\mathrm{e}^{-\kappa_1 t} + l_2\mathrm{e}^{-\alpha t}),$$

for $\forall i,j \in \mathcal{V}$. Thus, we obtain that $\lim_{t\to\infty}\bar{p}_i^e(t) - \bar{p}_j^e(t) = 0$ and $\lim_{t\to\infty} q_i(t) = 0$ exponentially. Since $s_i(t) = \dot{\bar{p}}_i^e(t) - q_i(t)$, one can further derive that $\bar{v}_i^e(t) = \dot{\bar{p}}_i^e(t) = q_i(t) + s_i(t)$. It then follows that

$$\|\bar{v}_i^e(t)\| \leq 2\gamma_i(l_1\mathrm{e}^{-\kappa_1 t} + l_2\mathrm{e}^{-\alpha t}) + s_i(0)\mathrm{e}^{-k_\theta t}, \tag{39}$$

which implies that $\bar{v}_i^e(t)$ converges to 0 exponentially. □

Proposition 3: For the auxiliary system (12b). if $\lim_{t\to\infty}\theta_i(t) = 0$, it then infers that $\lim_{t\to\infty}\eta_i(t) = \lim_{t\to\infty}\dot{\eta}_i(t) = 0$.

Proof: The detailed proofs refer to Lemma 1 in [40]. □

Theorem 1: Consider multiple UAVs modeled by system (1) under Assumption 1. The control objective (3) can be achieved under the hierarchical control strategies (12) and (19).

Proof: Note that $\lim_{t\to\infty}(\bar{p}_i^e(t) - \bar{p}_j^e(t)) = 0$ and $\lim_{t\to\infty}\bar{v}_i^e(t) = 0$, $i,j \in \mathcal{V}$ are shown in Proposition 2, and $\lim_{t\to\infty}\eta_i(t) = \lim_{t\to\infty}\dot{\eta}_i(t) = 0$, $i \in \mathcal{V}$ is given in Proposition 3. It then follows that $\lim_{t\to\infty}(p_i^e(t) - p_j^e(t)) = 0$ and $\lim_{t\to\infty} v_i^e(t) = 0$. Based on $\lim_{t\to\infty}\sigma_i^e(t) = 0$, $\lim_{t\to\infty}\omega_i^e(t) = 0$ in Proposition 1, we obtain that the control objective (3) can be achieved. □

Remark 3: The multiple UAVs formation control problem under switching topologies is also considered in [27]. Note that the algorithm of [27] does not guarantee a smooth control force due to the discontinuous input of the switching topologies. In contrast, our algorithm (12) addresses this problem and achieves smooth control force.

Remark 4: The distributed algorithm (12) is partly motivated by the works of [37] and [38]. Even though [37] and [38] addressed the discontinuity issues in control input, they are only applicable to the consensus problem for Euler-Lagrange (EL) systems with zero velocity. In contrast to [37] and [38], the developed formation control scheme (12) offers two advantages: (i) Our proposed algorithm (12) is applied to the quadrotor dynamic model that is a class of underactuated systems while EL systems in [37] and [38] is fully actuated. (ii) The developed algorithm (12) achieves dynamic formation control objective with any nonzero velocity while the results in [37] and [38] are restricted to static one.

We next show that the event-triggering updated rule (16) does not exhibit the Zeno behavior.

Theorem 2: Under Assumption 1, consider the error dynamic system (4) under the distributed algorithm (12). The Zeno behavior of event-triggering condition (16) can be excluded.

Proof: Define the state error $e_i(t) = \bar{p}_i(t) - \hat{\bar{p}}_i(t) - \int_{t_k^i}^t v_r(s)\mathrm{d}s$, and the event-triggering condition (16) can be written as

$$t_{k+1}^i = \left\{t \geq t_k^i \middle| \|e_i(t)\| - c_0\mathrm{e}^{-\alpha t} \geq 0\right\}. \tag{40}$$

To derive the interevent time, consider that the ith UAV triggers at instant $t_k^i \geq 0$. It then follows that $e_i(t_k^i) = 0$ and $\|e_i(t)\| - c_0\mathrm{e}^{-\alpha t} \leq 0$ during $[t_k^i, t_{k+1}^i)$.

We analyze the dynamics of $e_i(t)$ as follows:

$$\frac{\mathrm{d}\|e_i(t)\|}{\mathrm{d}t} = \frac{e_i^{\mathrm{T}}(t)\dot{e}_i(t)}{\|e_i(t)\|} \leq \|\dot{e}_i(t)\| = \|\dot{\bar{p}}_i^e\| = \|\bar{v}_i^e\|. \tag{41}$$

Substituting (39) into (41) yields

$$\frac{\mathrm{d}\|e_i(t)\|}{\mathrm{d}t} \leq 2\gamma_i(l_1\mathrm{e}^{-\kappa_1 t} + l_2\mathrm{e}^{-\alpha t}) + s_i(0)\mathrm{e}^{-k_\theta t}. \tag{42}$$

Performing some algebraic operations, we can derive the following result from (42)

$$\|e_i(t)\| \leq \left(2\gamma_i(l_1\mathrm{e}^{-\kappa_1 t_k^i} + l_2\mathrm{e}^{-\alpha t_k^i}) + s_i(0)\mathrm{e}^{-k_\theta t_k^i}\right)(t_{k+1}^i - t_k^i). \tag{43}$$

The next event will not be triggered until $\|e_i(t)\| = c_0\mathrm{e}^{-\alpha t}$, which implies that $\|e_i(t_{k+1}^i)\| = c_0\mathrm{e}^{-\alpha t_{k+1}^i}$ at the event time t_{k+1}^i. Based on (43), we obtain that

$$c_0\mathrm{e}^{-\alpha\psi} \leq (2\gamma_i l_1\mathrm{e}^{(\alpha-\kappa_1)t_k^i} + 2\gamma_i l_2 + s_i(0)\mathrm{e}^{(\alpha-k_\theta)t_k^i})\psi, \tag{44}$$

where $\psi = t_{k+1}^i - t_k^i$ is interevent time. For all $t_k^i \geq 0$ with $k_\theta \geq \alpha$ and $\kappa_1 > \alpha$, the solution ψ is greater or equal to $\psi_{\min}$ given

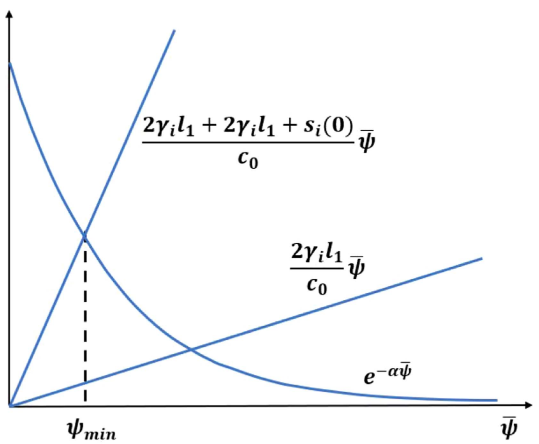

Fig. 2. The lower bound of interevent time $\psi_{\min}$.

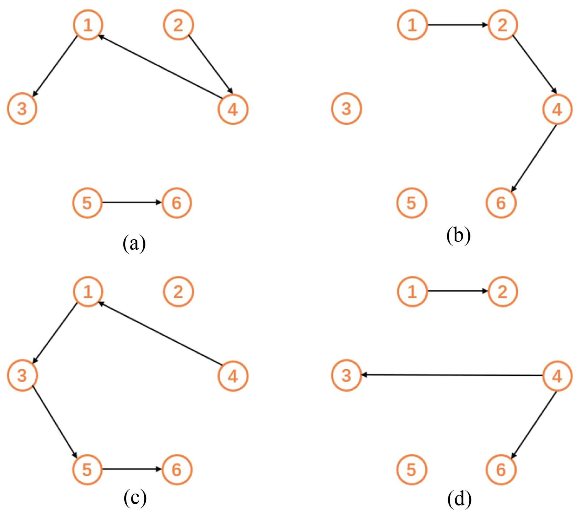

Fig. 3. Interaction topologies. (a) $\mathcal{G}_1$. (b) $\mathcal{G}_2$. (c) $\mathcal{G}_3$. (d)$\mathcal{G}_4$.

by

$$c_0 e^{\alpha\psi} = (2\gamma_i l_1 + 2\gamma_i l_2 + s_i(0))\psi,$$

which implies that ψ is strictly positive. This is visually illustrated in Fig. 2. The proof is completed. □

Remark 5: The reason for selecting parameters α and k_θ is to ensure the convergence rate of the state error e_i slower than the state $\tilde{\zeta}$ of (30), thereby excluding the Zeno behavior. The selection of control parameters depends on the information κ_1 associated with the switching topologies. Nevertheless, this value can be estimated using the result $\kappa_1 \geq 4/(N(N-1))$ in [41], where N is the number of nodes in the graph, which enables the exponential convergence of the virtual error.

V. Simulation

This section provides a simulation example to show the effectiveness of the developed control algorithms (12) and (19). Consider that a UAVs formation is composed of six quadcopter drones. Fig. 3 shows four possible interaction topologies among UAVs. The graph $\mathcal{G}(t)$ switches periodically at every one second in the order of $\{\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3, \mathcal{G}_4, \mathcal{G}_1, \cdots\}$. Obviously, the graph

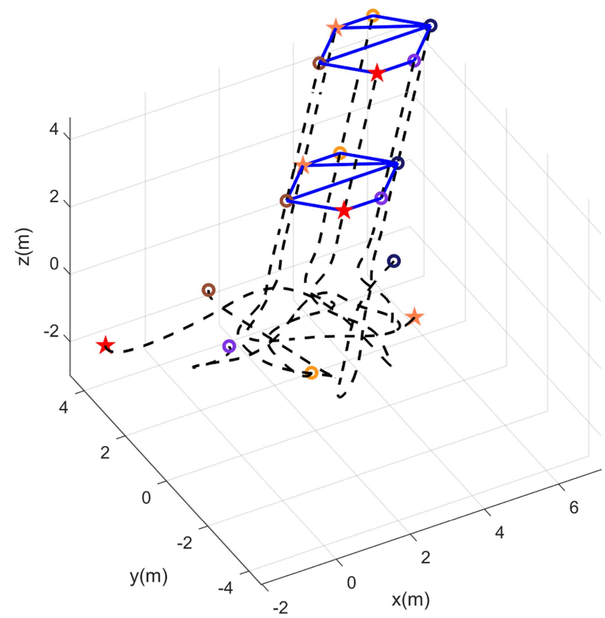

Fig. 4. Maneuver trajectory of the whole formation.

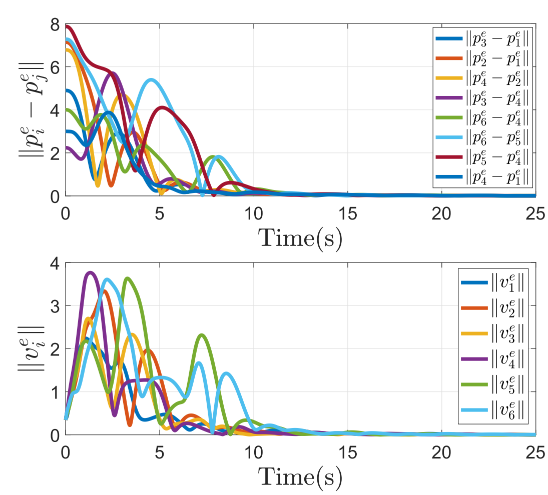

Fig. 5. Position offset errors $\|p_i^e - p_j^e\|$ and velocity errors $\|v_i - v_r\|$.

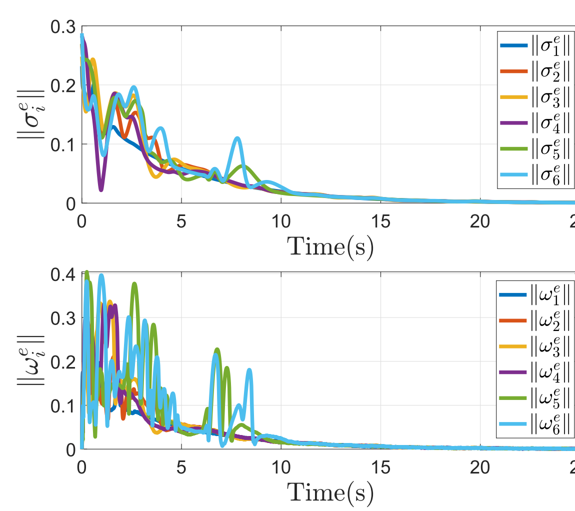

Fig. 6. Evolutions of $\|\sigma_i^e\|$ and $\|\omega_i^e\|$.

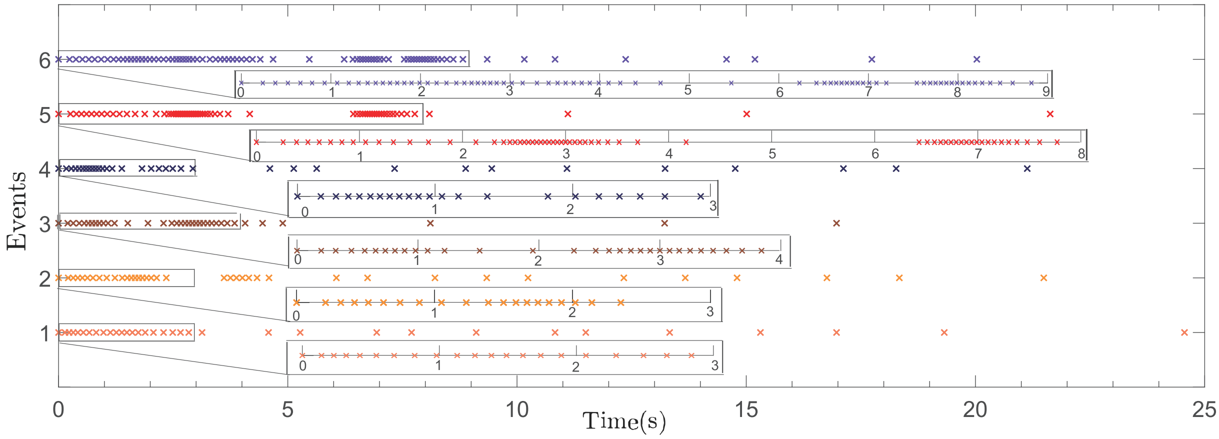

Fig. 7. Event-triggering instant sequences of the six UAVs.

$\mathcal{G}(t)$ does not contain a spanning tree for any $t \geq 0$ but the union of the four graphs has.

The relative positions of target formation are set to be: $\delta_{31} = \delta_{64} = [-1, -1, 0]^T$, $\delta_{42} = [1, -1, 0]^T$, $\delta_{21} = \delta_{65} = [1, 0, 0]^T$, $\delta_{34} = [-3, 0, 0]^T$, $\delta_{53} = [1, -1, 0]^T$, $\delta_{31} = [-2, 1, 0]^T$. The reference velocity is chosen as $v_r = [0.2, 0.2, 0.2]^T$ and inertia matrixes of UAVs are given as $J_i = i * 0.1 * [1, 0, 0.3; 0, 1.5, 1; 0.3, 0, 1]\text{kg}\cdot\text{m}^2$. The initial states $(p_i, v_i, \sigma_i, \omega_i)$ of the UAVs are selected as $p_1(0) = [3, -3, 2]^T$, $p_2(0) = [3, 2, -3]^T$, $p_3(0) = [-2, -2, 4]^T$, $p_4(0) = [3, -2, 3]^T$, $p_5(0) = [-2, 3, -1]^T$, $p_6(0) = [-2, -3, 3]^T$, $\sigma_i(0) = [-0.18, 0.09, -0.04]^T$, $i = 1, 3, 5$, $\sigma_i(0) = [0.12, -0.22, -0.22]^T$, $i = 2, 4, 6$.

The gain parameters in the control force (12) are chosen as follows: $k_\theta = 1$, $k_\eta = 2$, $\gamma_i = 5$, $k_1 = 5$, $k_2 = 8$, $\lambda_{h,i} = 5$, the parameters of $x_{h,i}$ are randomly selected in [0,1], for $h = 1, \ldots, 4$ and $i \in \mathcal{V}$, i.e., the order of the high-order dynamical system (13) is set to be $k = 4$. The parameters in the event-triggering mechanism (16) are designed as $c_0 = 0.3$ and $\alpha = 0.15$.

Fig. 4 depicts the motion of the whole formation with six UAVs at $t = 0s$, $t = 15s$, $t = 25s$ in the three-dimensional space, respectively. Fig. 5 exhibits the position offset errors $\|p_i^e - p_j^e\|$ and velocity errors $\|v_i - v_r\|$ among the UAVs. It can be observed that $\|p_i^e - p_j^e\|$ and $\|v_i - v_r\|$ both reach zero within $15s$ under the control force (12). From the results in Fig. 4 and Fig. 5, we observe that the developed formation algorithm (12) can achieve the formation control objective. Fig. 6 depicts that attitude error σ_i^e and angular velocity error ω_e^i of each UAV are driven to zero, which also demonstrate the effectiveness of the attitude control torque (19).

Fig. 7 shows the triggering times for each UAV under event-triggering mechanism (16). In particular, we calculate the number of triggering events for all the UAVs as $n_1 = 36, n_2 = 46, n_3 = 36, n_4 = 39, n_5 = 63$, $n_6 = 81$. Moreover, if the formation algorithm (12) is implemented with periodic communication, one derive that the communication number of periodical manners is $n = 250$ under the fixed sampling step-size $T = 0.1$. Thus, the reduced percentage of communication transmission for each UAV can be obtained as $(n - n_i)/n \times 100\%$, that is $85.6\%, 81.6\%, 85.6\%, 84.4\%, 74.8\%, 67.6\%$. Based on the above results, we conclude that the developed event-triggered formation algorithm reduces the communication costs.

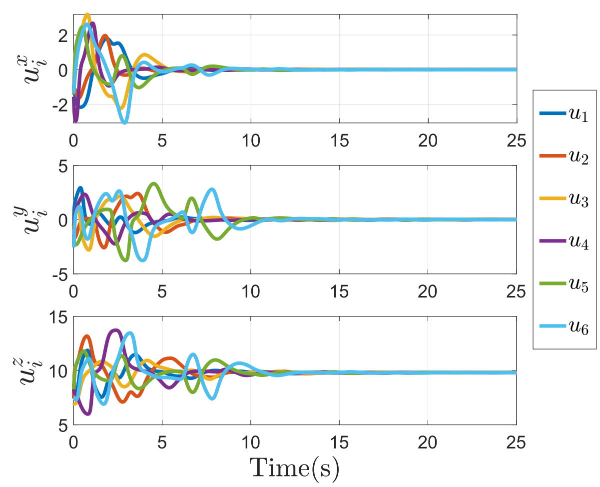

Fig. 8. The formation control force u_i of each UAV.

We further demonstrate that the developed formation controller u_i remains continuous and even smooth despite the discontinuities caused by the event-triggering mechanism and topology switching. Fig. 8 describes the response curves of the control force u_i of UAVs, from which we observe that all three elements of the formation control input $u_i, \forall i \in \mathcal{V}$ are all continuous. To emphasize smooth characteristics of our developed formation algorithm (12), we compare it with the formation controller proposed in [27]. Fig. 9 depicts the derivative of the formation control input, $\dot{u}_i$. From this, it is evident that all the elements of $\dot{u}_i$ are continuous. Fig. 10 describes the derivative $\dot{u}_i$ for the controller in [27], from which we observe that the derivative $\dot{u}_i$ are discontinuous at many moments. Based on the results in Fig. 9 and Fig. 10, it is clear that the developed formation algorithm achieves smooth control performance, which effectively addresses the discontinuity issues caused by switching topologies and event-triggering mechanism.

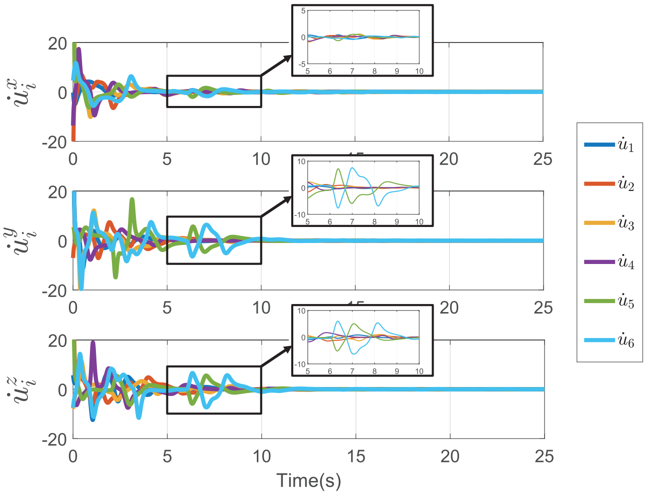

Fig. 9. The derivative $\dot{u}_i$ for the formation control algorithm (12).

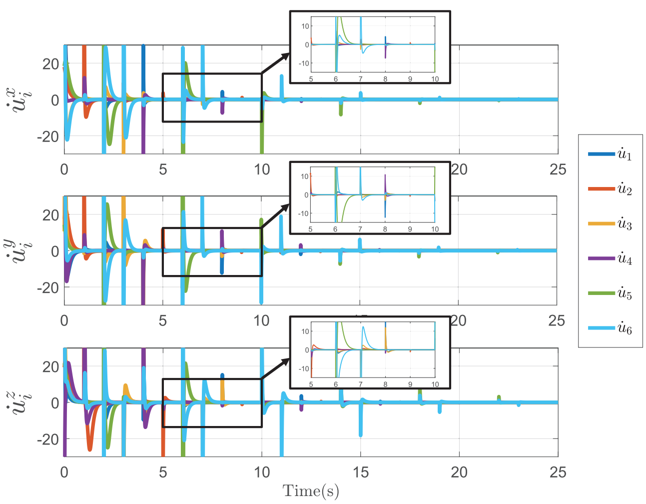

Fig. 10. The derivative $\dot{u}_i$ for the formation controller proposed in [27].

VI. Conclusion

This paper delves into the multiple UAVs formation control problem under switching directed topologies. Taking into account the underactuated dynamics of quadrotors, a hierarchical control framework comprising position loop and attitude loop is introduced. A distributed control force, which only relies on relative position information from neighbors and reference velocity information, is proposed. This nice property makes it easier to be implemented in practical applications. By using a high-order dynamical system, the discontinuity issues caused by event-triggering mechanism and switching topologies is addressed, thereby achieving a smooth control force for each UAV. The introduced event-triggering mechanism also exhibits effectiveness in reducing communication cost.

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